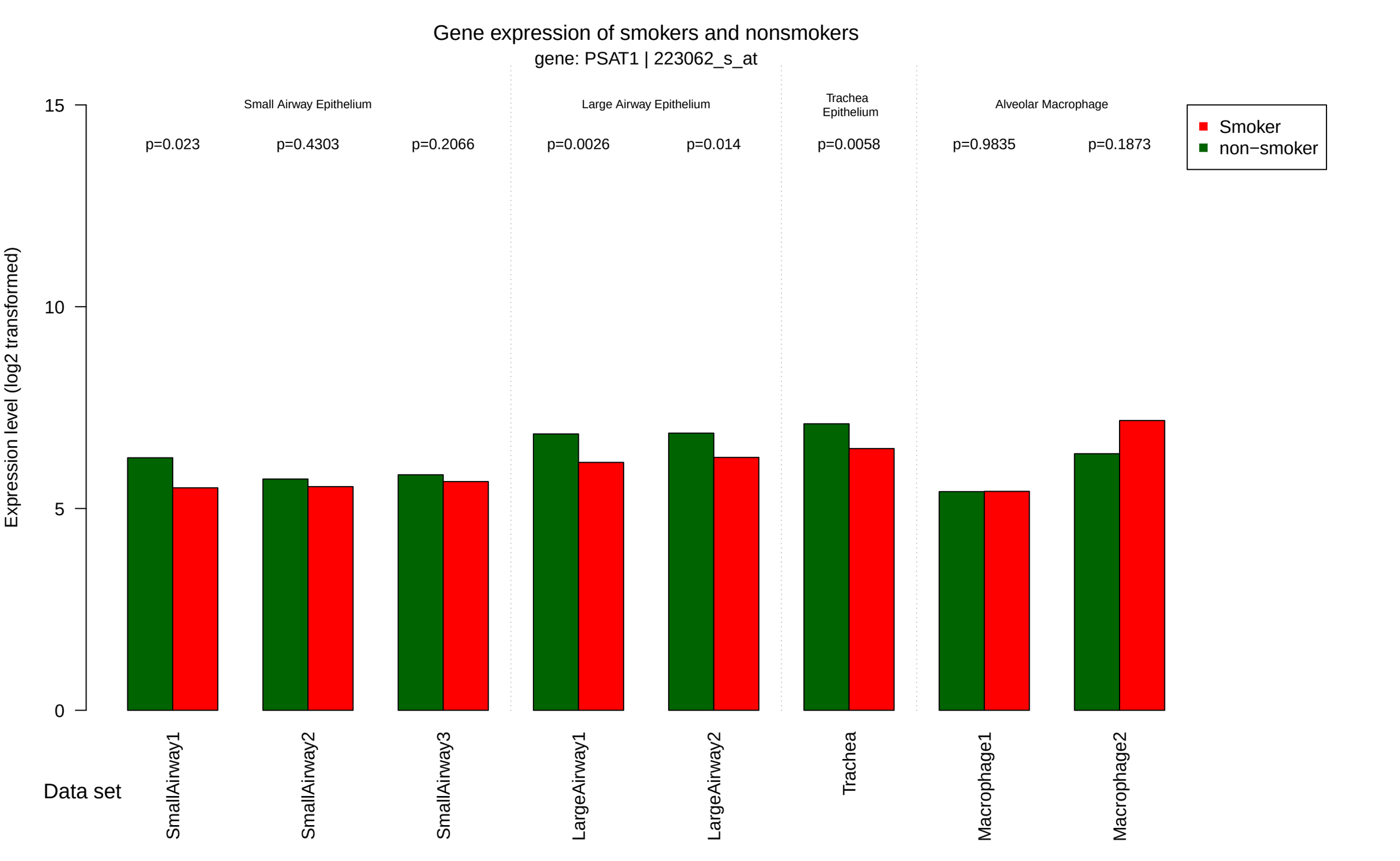
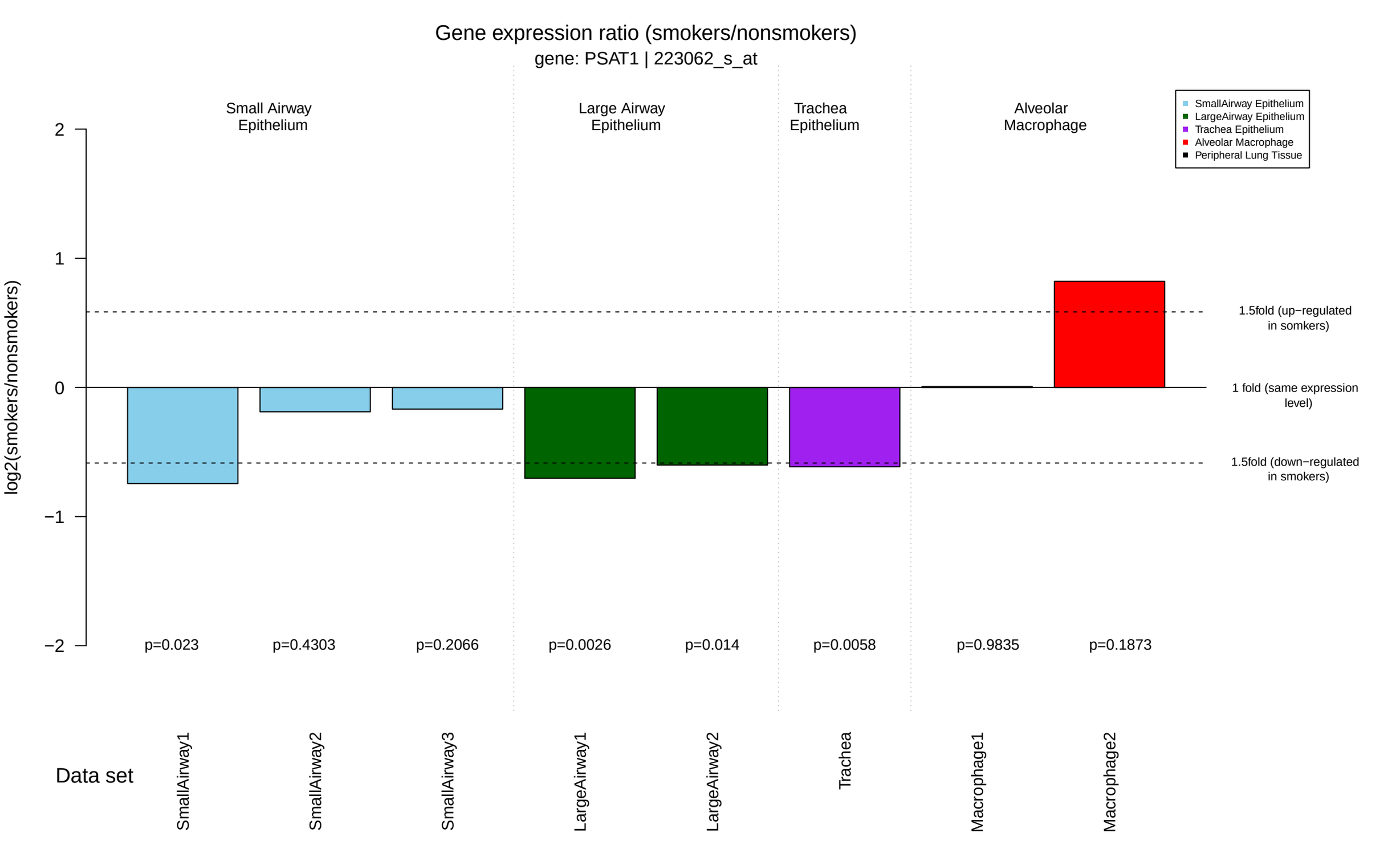
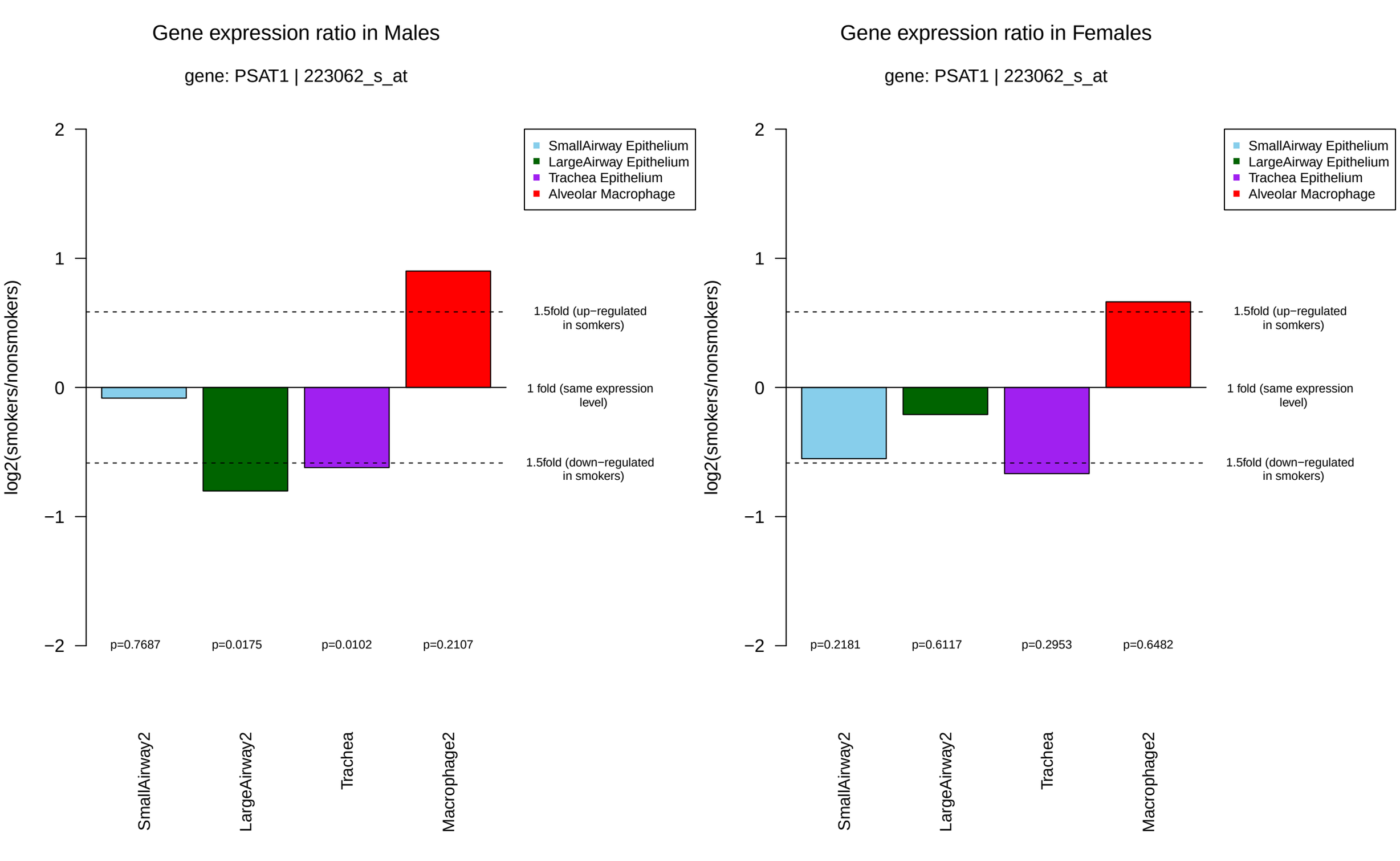
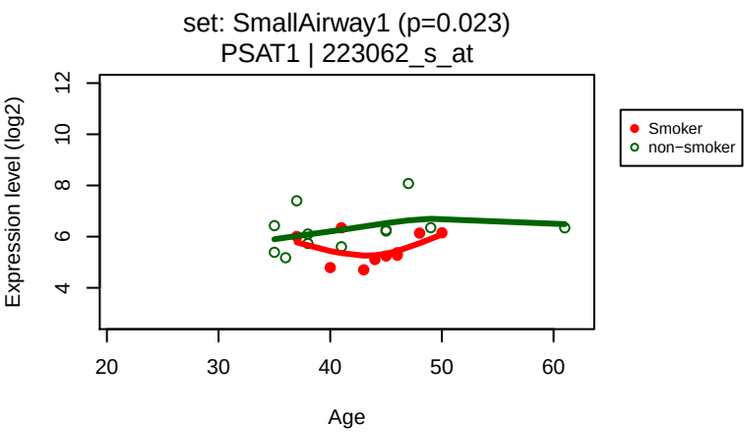
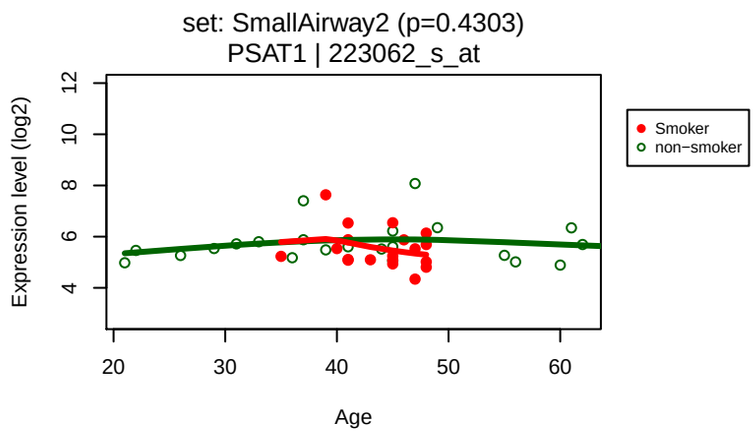
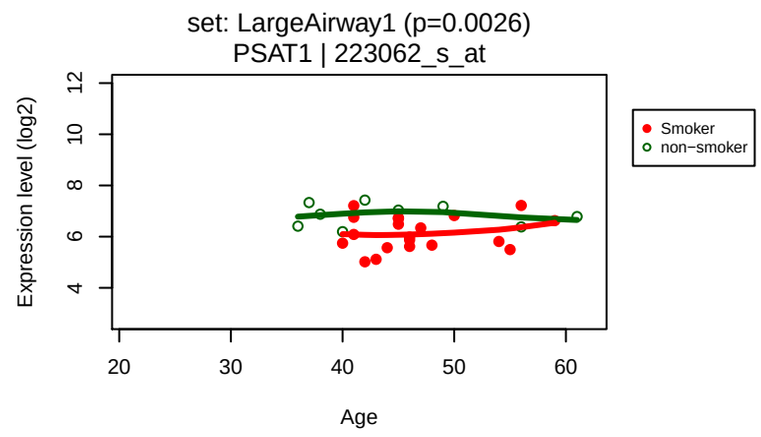

Gene expression of smokers and nonsmokers
gene: PSAT1 | 223062_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.023
p=0.4303
p=0.2066
p=0.0026
p=0.014
p=0.0058
p=0.9835
p=0.1873
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

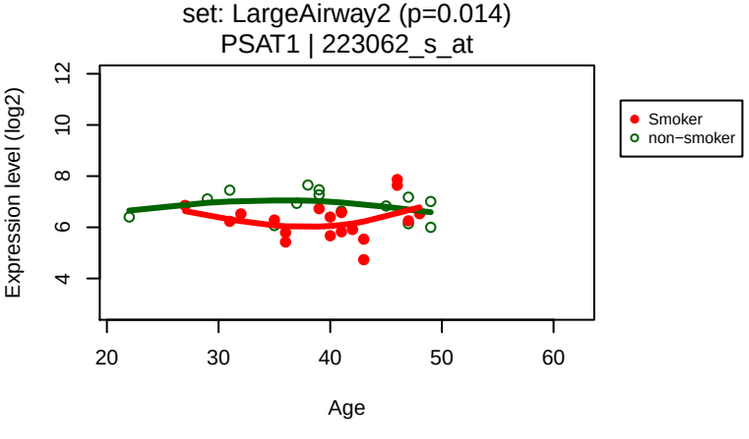

Gene expression ratio (smokers/nonsmokers)
gene: PSAT1 | 223062_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.023
p=0.4303
p=0.2066
p=0.0026
p=0.014
p=0.0058
p=0.9835
p=0.1873
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

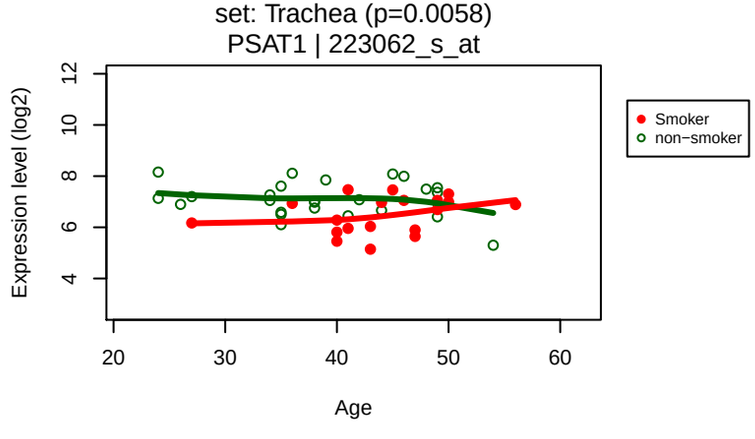

Gene expression ratio in Males
gene: PSAT1 | 223062_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7687
p=0.0175
p=0.0102
p=0.2107
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PSAT1 | 223062_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2181
p=0.6117
p=0.2953
p=0.6482
SmallAirway2
LargeAirway2
Trachea
Macrophage2

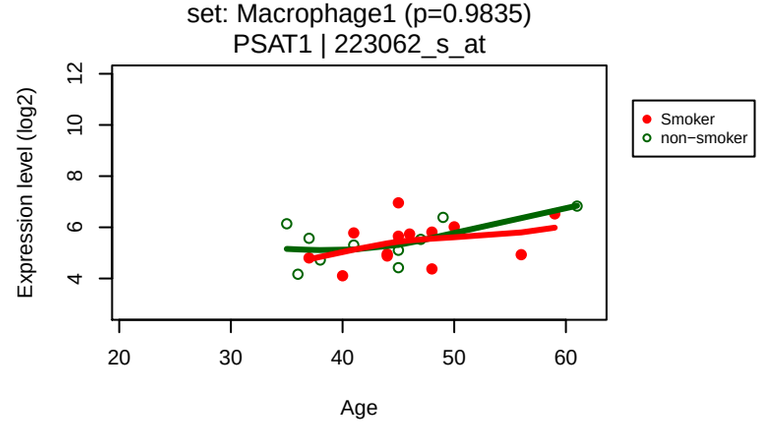

set: SmallAirway1 (p=0.023)
PSAT1 | 223062_s_at
Expression level (log2)
Age
Smoker
non−smoker

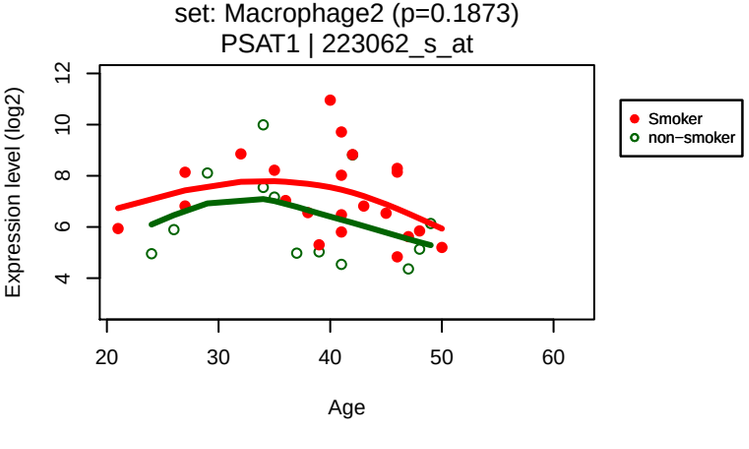

set: SmallAirway2 (p=0.4303)
PSAT1 | 223062_s_at
Expression level (log2)
Age
Smoker
non−smoker

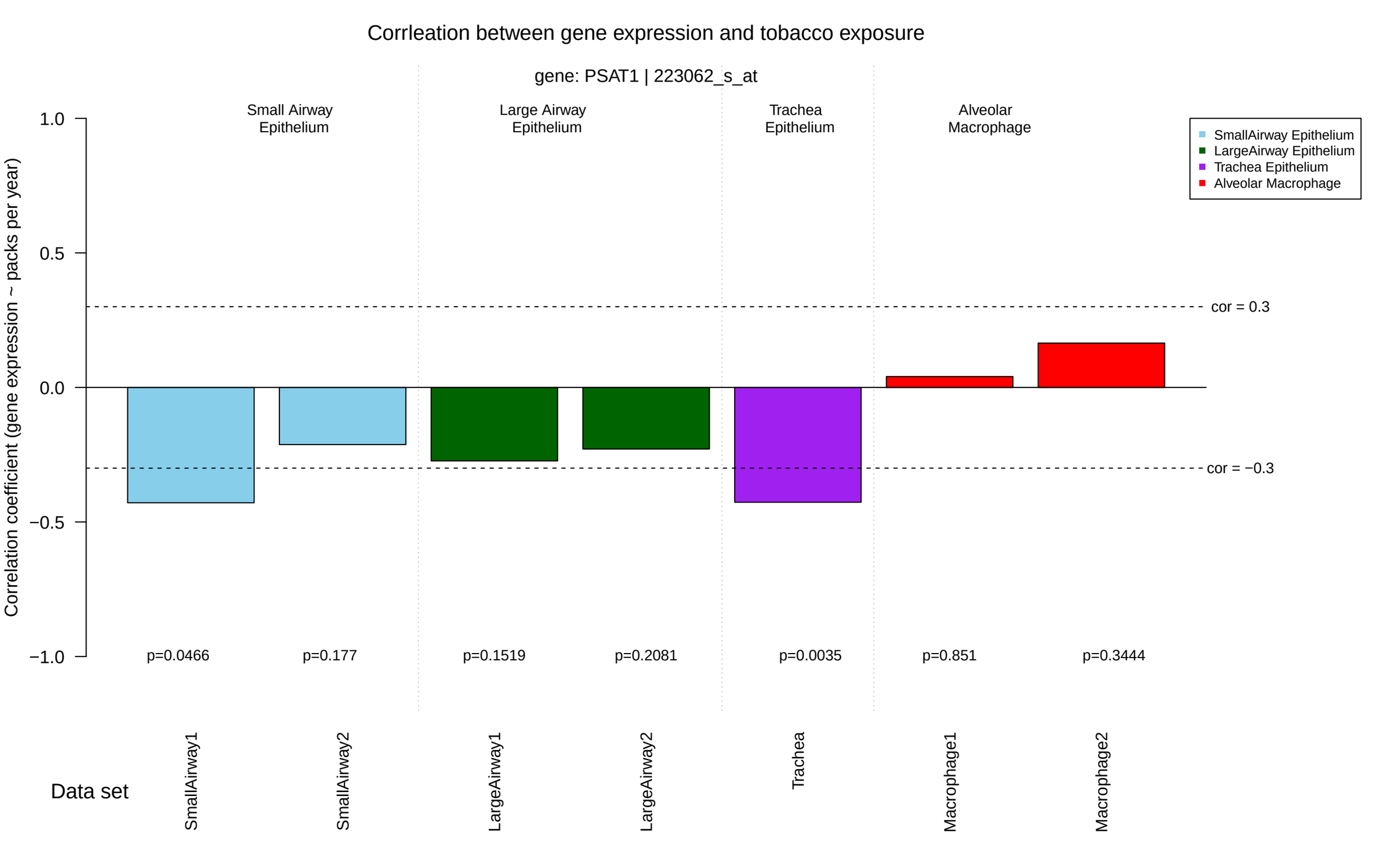

set: LargeAirway1 (p=0.0026)
PSAT1 | 223062_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.014)
PSAT1 | 223062_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0058)
PSAT1 | 223062_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9835)
PSAT1 | 223062_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1873)
PSAT1 | 223062_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PSAT1 | 223062_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0466
p=0.177
p=0.1519
p=0.2081
p=0.0035
p=0.851
p=0.3444
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2